



LOST BOYS CYBER

THREAT INTELLIGENCE • MALWARE ANALYSIS • DETECTION ENGINEERING



Unit 42 Direct-to-IP Malware Communications Daily Detection Review Update

Threat Report

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AI Generation: This report was generated with AI assistance and is provided for informational purposes only. All findings, IOCs, detection rules, hunting queries, attribution assessments, and recommendations must be independently reviewed and validated by a qualified security professional before operational use.

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Unit 42 Direct-to-IP Malware Communications Daily Detection Review Update

1. Daily Update Scope

This 2026-08-13 automation run selected this item for daily analyst review based on current or recently updated public reporting. Where sources are older than 72 hours, the report explicitly treats the source window as expanded to seven days because the topic remains operationally relevant.

2. Executive Summary

Unit 42 reported that a substantial share of malware samples communicate directly with IP addresses and bypass DNS telemetry. This matters for defenders that over-index on DNS controls and do not hunt rare direct-to-IP egress from endpoints, servers, and cloud workloads. This report converts public reporting into immediate defender actions, detection logic, and threat-hunting queries. Claims are limited to the cited sources and Lost Boys Cyber analytic implications.

3. Intelligence Requirements

Question	Current Answer	Confidence
Why review this today?	It is current or recently updated and has practical exposure across monitored environments.	Medium
Which teams should act?	SOC, identity, endpoint, cloud/SaaS, vulnerability management, and business-system owners.	Medium
What evidence should be collected first?	Endpoint process/network telemetry, identity sign-ins, admin audit logs, proxy/DNS logs, and third-party access logs.	High

4. Source Review & Confidence

Source	Contribution	Confidence
Unit 42 direct-to-IP malware research	Primary malware communications research.	Medium-High
CISA malware analysis and network defense resources	Government malware-analysis context.	Medium-High
MITRE ATT&CK Command and Control tactics	TTP mapping context.	Medium-High

5. Threat Activity Overview

Unit 42 reported that a substantial share of malware samples communicate directly with IP addresses and bypass DNS telemetry. This matters for defenders that over-index on DNS controls and do not hunt rare direct-to-IP egress from endpoints, servers, and cloud workloads.

Activity Element	Analyst Assessment	Collection Priority
Initial access	May involve exploitation, social engineering, identity abuse, or supplier	High

	compromise depending on environment.	
Execution / foothold	Look for script execution, unusual child processes, package hooks, API abuse, or living-off-the-land tools.	High
Command and control	Monitor rare direct-to-IP, unusual TLS destinations, SaaS/API anomalies, and abnormal OAuth/token flows.	Medium
Impact	Data theft, extortion, service disruption, or supply-chain propagation are plausible business impacts.	Medium

6. Affected Sectors and Exposure

Sector / Environment	Relevance	Priority Checks
Enterprise IT	Identity, endpoints, SaaS, and perimeter services are common attack surfaces.	MFA posture, EDR coverage, admin logs
Managed service / supply chain	Trusted relationships amplify blast radius.	Vendor access inventory, RMM restrictions, provenance
Critical operations	Outages may become operational disruptions even when OT is segmented.	BCP playbooks, segmentation, monitoring

7. Tactics, Techniques, and Procedures

Tactic / Phase	Observed or Plausible Technique	Evidence / Rationale
Initial Access	T1190 Exploit Public-Facing Application / T1566 Phishing / T1195 Supply Chain Compromise	Current reporting emphasizes exploitation, social engineering, or package/supplier paths.
Execution	T1059 Command and Scripting Interpreter	Post-compromise actions often require scripts, installers, or living-off-the-land execution.
Persistence	T1098 Account Manipulation	Account/token changes are common in identity-heavy intrusions.
Command and Control	T1105 Ingress Tool Transfer / T1090 Proxy	Payload retrieval or proxying may follow initial foothold.
Impact	T1486 Data Encrypted for Impact / T1041 Exfiltration Over C2 Channel	Ransomware/extortion and data theft remain common outcomes.

8. Infrastructure and Indicators

Hard IOCs were not consistently available across all public sources reviewed for this daily item. The practical observable set is behavioral.

Observable	Type	Operational Use
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Unexpected package install, script, or automation job	Behavioral	Supply-chain and execution triage
Unusual direct-to-IP outbound traffic	Network behavior	Malware C2 hunting
New OAuth grant, token, or admin session	Identity behavior	Account takeover triage
RMM/tooling execution outside maintenance window	Endpoint behavior	Intrusion/ransomware precursor

9. Detection Engineering

9.1. Detection Summary

Signal	Telemetry Source	Analytic Goal
Suspicious tool/package execution	DeviceProcessEvents / Sysmon	Find initial execution and staging
Direct-to-IP or rare outbound connections	DeviceNetworkEvents / firewall	Find C2 or staging without DNS
Identity and SaaS access anomalies	SigninLogs / audit logs	Find trusted-relationship abuse
<pre> title: Unit 42 Direct-to-IP Malware Communications Daily Detection Review Update Daily Priority Signals id: unit-42-direct-to-ip-malware-communications-daily-detection-review-update-20260813 status: experimental logsource: product: windows category: process_creation detection: selection_terms: CommandLine contains: 'direct ip' CommandLine contains: 'hard-coded ip' CommandLine contains: 'p2p' CommandLine contains: 'beacon' CommandLine contains: 'RemoteIP' condition: selection_terms level: medium </pre>		

10. Threat Hunting

10.1. Hunt Hypothesis

If this activity is present, analysts should find one or more of: suspicious execution tied to the topic keywords, outbound communications to rare infrastructure, or identity events inconsistent with normal administrative patterns.

<pre> let lookback = 14d; DeviceProcessEvents where Timestamp > ago(lookback) where ProcessCommandLine has_any ('direct ip', 'hard-coded ip', 'p2p', 'beacon', 'RemoteIP') or FileName has_any ('direct ip', 'hard-coded ip', 'p2p', 'beacon', 'RemoteIP') project Timestamp, DeviceName, AccountName, FileName, ProcessCommandLine, InitiatingProcessFileName union (DeviceNetworkEvents where Timestamp > ago(lookback) where RemoteUrl has_any ('direct ip', 'hard-coded ip', 'p2p', 'beacon', 'RemoteIP') or InitiatingProcessCommandLine has_any ('direct ip', 'hard-coded ip', 'p2p', 'beacon', 'RemoteIP') or RemoteIPType == 'Public' project Timestamp, DeviceName, AccountName=InitiatingProcessAccountName, FileName=InitiatingProcessFileName, ProcessCommandLine=strcat(RemoteUrl, ':', RemotePort), InitiatingProcessFileName) SigninLogs where TimeGenerated > ago(14d) where AppDisplayName has_any ('direct ip', 'hard-coded ip', 'p2p', 'beacon', 'RemoteIP') or UserAgent has_any ('direct ip', 'hard-coded ip', 'p2p', 'beacon', 'RemoteIP') </pre>
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```
| summarize count(), makeset(IPAddress, 10), makeset(AppDisplayName, 10) by UserPrincipalName,
bin(TimeGenerated, 1d)
```

11. Risk Assessment

Risk Dimension	Assessment	Rationale
Likelihood	Medium	Topic is current and broadly applicable, but local exposure varies.
Impact	High	Identity, supply-chain, extortion, or operational dependencies can create material disruption.
Detection Confidence	Medium	Behavioral analytics require tuning and source-specific enrichment.

12. Recommended Actions

Priority	Owner	Action
Critical	SOC	Run the included hunts for the last 14 days and review high-confidence hits.
High	Identity / SaaS	Review OAuth grants, privileged sessions, and anomalous admin changes.
High	Endpoint / Network	Monitor rare direct-to-IP egress, script execution, and RMM/tool misuse.
Medium	Business Owner	Validate continuity and third-party escalation paths for impacted workflows.

13. References

- [1] Unit 42 direct-to-IP malware research: <https://unit42.paloaltonetworks.com/malware-bypass-dns-direct-to-ip/>
- [2] CISA malware analysis and network defense resources: <https://www.cisa.gov/resources-tools/resources/malware-analysis-reports>
- [3] MITRE ATT&CK Command and Control tactics: <https://attack.mitre.org/tactics/TA0011/>